

Integrity Concerns -- Guided Notes ANSWER KEY

Unit tp-6.2 | CompTIA Network+ N10-009 Obj. FC0-U71 6.1 | N10-008 FC0-U71 6.1

For instructor use / distribute after students complete the notes.

Question 1

Integrity in security means that data is _____, _____, and has not been _____ without authorization. An integrity violation occurs when data is modified by someone who was not _____ to make that change -- whether by an attacker, a system error, or accidental editing.

Answer: Accurate; complete; altered (changed); authorized.

Question 2

Hashing is the primary technical control for verifying integrity. A hash function produces a fixed-length _____ (digest) from input data. If even _____ bit of the data changes, the hash output changes completely. To verify integrity, compare the _____ hash to the _____ hash after transmission or storage.

Answer: Hash value (fingerprint); one; received; original.

Question 3

SHA-256 (Secure Hash Algorithm 256-bit) is a widely used cryptographic hash function. It is _____ -way -- you cannot reverse the hash to recover the original input. MD5 produces a 128-bit hash but is considered _____ for security purposes because collisions can be engineered. A collision is when two _____ inputs produce the _____ hash output.

Answer: One; weak (cryptographically broken); different; same.

Question 4

A digital _____ ensures both integrity and _____ of a message or document. The sender hashes the message and _____ the hash with their private key. The receiver decrypts the hash with the sender's _____ key and re-hashes the message -- if the hashes match, the message is unmodified and the sender is authenticated.

Answer: Signature; authentication (non-repudiation); encrypts; public.

Question 5

Threats to integrity include: _____ injection (attackers insert malicious data into a system), _____ -in-the-middle attacks (attacker modifies data during transmission), and _____ (attackers redirect users to fraudulent websites by corrupting DNS records). All three result in users receiving _____ data.

Answer: SQL/code; man; DNS poisoning (cache poisoning); tampered (falsified).

Question 6

File integrity monitoring (FIM) tools alert administrators when critical system files are _____ modified. They do this by storing _____ of key files at baseline and comparing regularly. Examples include

AIDE (Linux) and Tripwire. FIM is especially important on _____ and _____ servers where unauthorized changes could indicate a compromise.

Answer: Unexpectedly; hashes; web; domain controller (critical infrastructure).

Question 7

_____ controls like audit logs and version control help maintain and demonstrate integrity. A _____ trail records every change to data including who made it, when, and what changed. This is essential for _____ investigations and regulatory _____ (HIPAA, PCI-DSS require audit logging).

Answer: Detective; audit; forensic; compliance.

Question 8 -- Real World Application

REAL WORLD: A bank publishes software update files on their website so customers can download the latest mobile app. An attacker compromises the web server and replaces the legitimate installer with a malware-laced version. The file is the same size and has the same name. How would publishing a SHA-256 hash prevent customers from installing the malicious file?

Answer: The bank should publish the SHA-256 hash of the legitimate installer alongside the download link (e.g., 'SHA-256: a3f9...'). Before installing, customers (or their software manager) run SHA-256 on the downloaded file. Even though the malicious version has the same filename and similar file size, its content is different from the original installer. A single-byte change causes the SHA-256 output to change completely (avalanche effect). When the customer computes the hash, it will not match the published value -- this is an immediate signal that the file has been tampered with. The customer should discard the file and alert the bank. This verification method is called hash-based integrity checking. Automated software management tools (package managers like apt, brew, or npm) do this automatically for every download, which is why supply chain attacks through package managers are difficult to execute undetected.